Q1. a) Bookwork , b) Bookwork.

c) Proof: For each $n \in \mathbb{N}$ it follows that

$$\frac{n}{n+2} < \frac{n}{n} = 1.$$

Hence, 1 is an upper bound for A. Let (a_n) be a sequence with $a_n = \frac{n}{n+2}$. Then since $\lim(a_n) = 1$ it follows that $\sup(A) = 1$. Also, for each $n \in \mathbb{N}$, we have that

$$\frac{n}{n+2} > \frac{n}{n+2n} = \frac{1}{3}.$$

That is, $\frac{1}{3}$ is a lower bound for A. Since, $\frac{1}{3} = \frac{1}{1+2} \in A$ it follows that $\frac{1}{3}$ must be the greatest lower bound for A. That is, $\inf(A) = \frac{1}{3}$. Finally, as $1 \notin A$, the maximum of A does not exist, but, $\inf(A) = \min(A) = \frac{1}{3}$. ☐

d) $\mathbb{I}$ does not form a complete ordered field since it is not closed under addition. Clearly $\sqrt{2} \in \mathbb{I}$ and $-\sqrt{2} \in \mathbb{I}$ yet $\sqrt{2} + (-\sqrt{2}) = 0 \notin \mathbb{I}$.

2. a) Bookwork. b) Bookwork

c) Scratch Work

Guess that $a_n \to 5/4$.

$$|a_n - 5/4| = \left|\frac{5n}{4n+4} - \frac{5}{4}\right|$$

$$= \left|\frac{-20}{16(n+1)}\right|$$

$$= \frac{5}{4(n+1)} < \frac{5}{4n} < \frac{5}{n}.$$

Proof: Given $\varepsilon > 0$, by the Archimedean property, there exists $N \in \mathbb{N}$ such that $5/N < \varepsilon$. Then, for every $n > N$, we have that

$$|a_n - 5/4| < 5/n < 5/N < \varepsilon. \quad \square$$

Scratch Work: Guess $b_n \to 1/3$.

$$|b_n - 1/3| = \left|\frac{2n^2 - n}{6n^2 - 3} - 1/3\right| = \left|\frac{3(2n^2-n) - (6n^2-3)}{3(6n^2-3)}\right|$$

$$< \left|\frac{-3n+3}{6n^2-3}\right| < \frac{3n}{|6n^2 - 3n^2|} = \frac{1}{n}.$$

Proof: Given $\varepsilon > 0$, by the Archimedean property, there exists $N \in \mathbb{N}$ such that $1/N < \varepsilon$. Hence, for every $n > N$, we have that

$$|b_n - 1/3| < 1/n < 1/N < \varepsilon. \quad \square$$

d) This is not true. Consider the sequence (a_n) with $a_n = (-1)^n/n$. Then $a_n \to 0$ but it is not monotone.

3. a) Bookwork b) Bookwork.

c) Proof: Let $a_k = \dfrac{2\sin(k) + \cos(k)}{k^7}$.

Then $|a_k| = \left|\dfrac{2\sin(k)+\cos(k)}{k^7}\right| \leq \left|\dfrac{2\sin(k)}{k^7}\right| + \left|\dfrac{\cos(k)}{k^7}\right|$

$$\leq \frac{2}{k^7} + \frac{1}{k^7} = \frac{3}{k^7}.$$

By the P-series test, absolute convergence test and comparison test, it follows that

$$\sum_{k=1}^{\infty} \frac{2\sin(k) + \cos(k)}{k^7},$$

is convergent.

d) Proof: Let $a_k = \dfrac{(-1)^k\sqrt{k^2+4}}{k^2+5k+4}$. Then, letting $b_k =$

have that $\lim\left(\dfrac{|a_k|}{b_k}\right) = \lim\left(\dfrac{k\sqrt{k^2+4}}{k^2+5k+4}\right) = 1$.

Since, $\sum_{k=1}^{\infty} 1/k$ diverges it follows that $\sum_{k=1}^{\infty} |a_k|$ diverges.

Now, $\lim \left(\frac{\sqrt{k^2+4}}{k^2+5k+4} \right) = \lim \left(\frac{\sqrt{1/k^2+4/k^4}}{1+5/k+4/k^2} \right) = 0.$

Define $f: [0,\infty) \to \mathbb{R}$ by $f(x) = \frac{\sqrt{x^2+4}}{x^2+5x+4}$.

Both $\sqrt{x^2+4}$ and x^2+5x+4 are differentiable on $[0,\infty)$ and thus f is differentiable on $[0,\infty)$. The derivative is given by

$$f'(x) = \frac{x(x^2+4)^{-1/2}(x^2+5x+4) - (2x+5)(x^2+4)^{1/2}}{(x^2+5x+4)^2}$$

$$= \frac{(x^3+5x^2+4x) - (2x+5)(x^2+4)}{(x^2+4)^{1/2}(x^2+5x+4)^2}$$

$$= \frac{(x^3+5x^2+4x) - (2x^3+5x^2+8x+20)}{(x^2+4)^{1/2}(x^2+5x+4)^2}$$

$$= \frac{-x^3-4x-20}{(x^2+4)^{1/2}(x^2+5x+4)^2}.$$

So, $f'(x) < 0$ for all $x \in [0,\infty)$. That is, the function is decreasing on $[0,\infty)$. Hence, $|a_{k+1}| < |a_k|$ for all $k \in \mathbb{N}$. By the alternating series test, it follows that

$$\sum_{k=1}^{\infty} \frac{(-1)^k \sqrt{k^2+4}}{k^2+5k+4}$$

is conditionally convergent. ∎

4. a) Bookwork b) Bookwork

c) Scratch Work : $f(2) = 4+2-2 = 4$.

So, $|f(x)-f(2)| = |x^2+x-6|$

$= |x+3||x-2|$.

If $|x-2|<1$ then $-1<x-2<1$

$\Rightarrow 1<x<3$

$\Rightarrow 4<x+3<6$.

Proof : Given $\varepsilon>0$, let $\delta = \min(\varepsilon/6, 1)$. Then, if $|x-2|<\delta$ it follows that

$|f(x)-f(2)| = |x+3||x-2|$

$< \delta|x+3|$

$\leq \varepsilon/6\,|x+3|$

$< \varepsilon/6 \cdot 6$

$= \varepsilon$. ∎

d) If n is odd, then $\lim_{x\to+\infty}(p(x)) = +\infty$ and $\lim_{x\to-\infty}(p(x)) = -\infty$. Hence, there exist $a, b \in \mathbb{R}$ with $p(a)<0$ and $p(b)>0$. Since, p is continuous on $\mathbb{R}$ is is continuous on $[a,b]$. By the intermediate value theorem, there exists

$c \in (a,b)$ such that $p(c)=0$. Hence, p has at least one real root.

5. a) Bookwork. b) Bookwork.

c) Let $a_k = \frac{(3k)!}{(k!)^3}$. Then,

$$\lim \left| \frac{a_{k+1}}{a_k} \right| = \lim \left| \frac{(3k+3)!}{((k+1)!)^3} \cdot \frac{(k!)^3}{(3k)!} \right|$$

$$= \lim \left| \frac{(3k+1)(3k+2)(3k+3)}{(k+1)^3} \right|$$

$$= 27.$$

Thus, $R = 1/27$.

d) This is false. Consider $f: \mathbb{R} \to \mathbb{R}$

$$f(x) = \begin{cases} x^2 \sin(1/x) & : x \neq 0 \\ 0 & : x = 0 \end{cases}.$$

Now, $f'(0) = \lim_{x \to 0} \left(\frac{x^2 \sin(1/x)}{x} \right) = 0$. So f is differentiable at zero. Additionally since x^2 and $\sin(1/x)$ is differentiable on $\mathbb{R} - \{0\}$ it follows that f is differentiable

on $\mathbb{R}$. Now, $f'(x) = 2x\sin(1/x) - \cos(1/x)$.

We have that $$f''(0) = \lim_{x \to 0} \left(\frac{2x\sin(1/x) - \cos(1/x)}{x} \right)$$

$$= \lim_{x \to 0} \left(2\sin(1/x) - \frac{\cos(1/x)}{x} \right).$$

This limit does not exist and hence f is not second differentiable at zero.